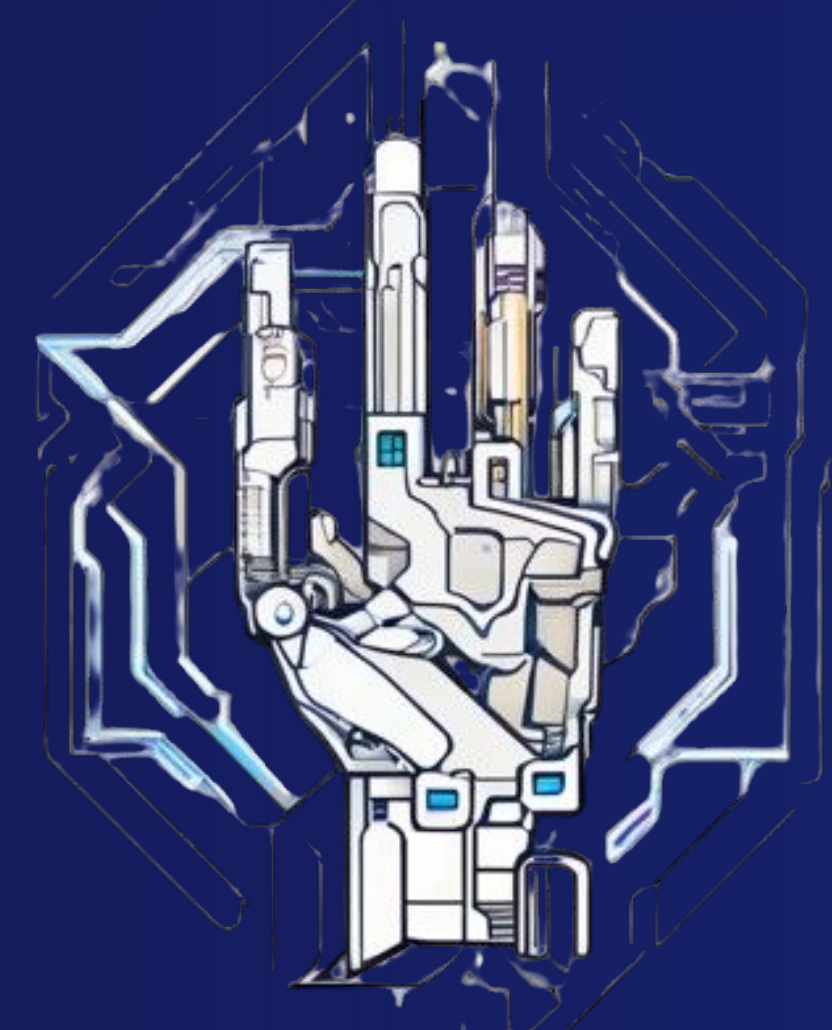


SignScribe

Robotic hand that translates live speech to ASL movements.



A. JAMES CLARK
SCHOOL OF ENGINEERING
Cyber-Physical Systems Engineering

AT

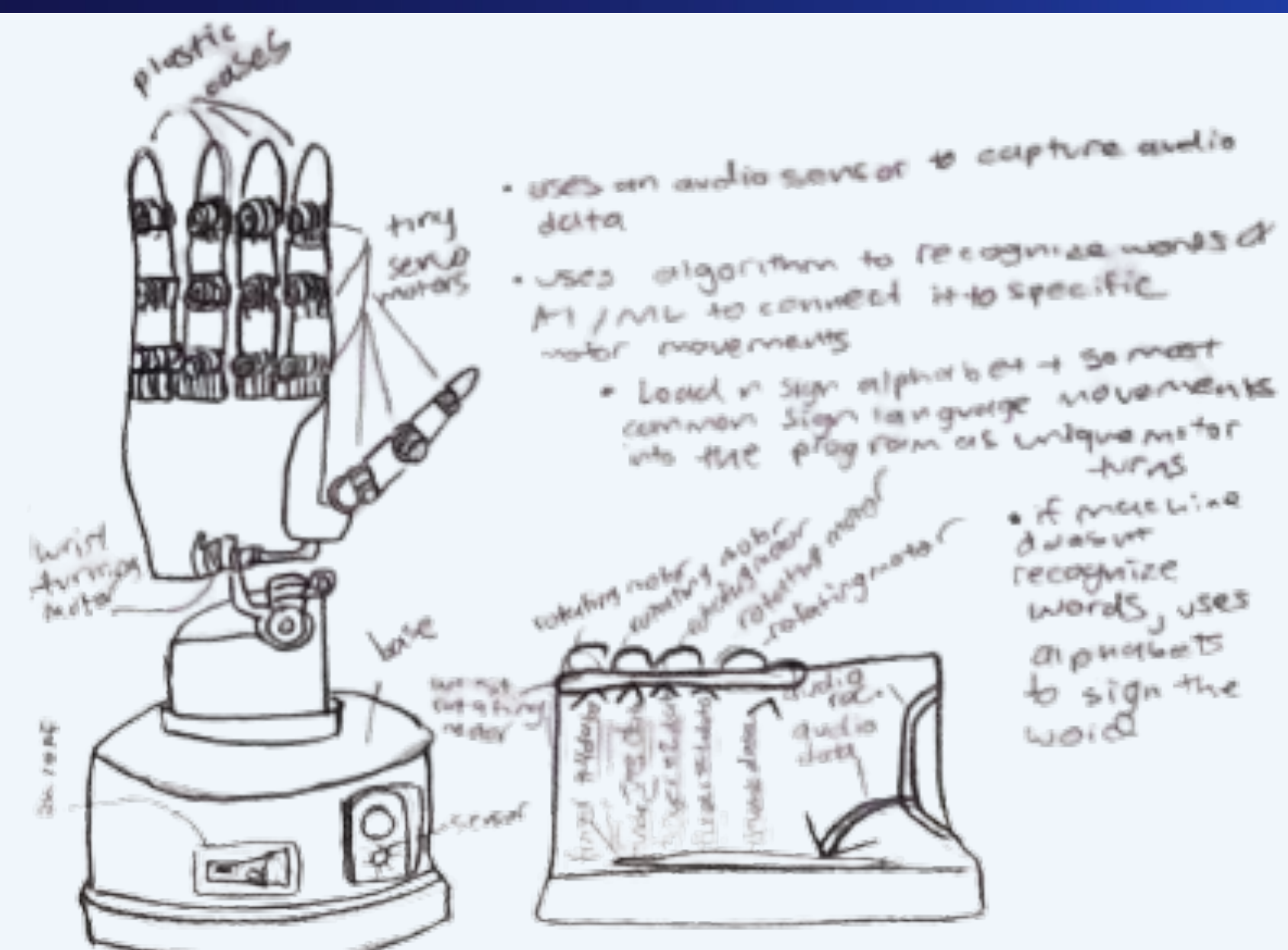


The Universities
AT SHADY GROVE

THE PROBLEM

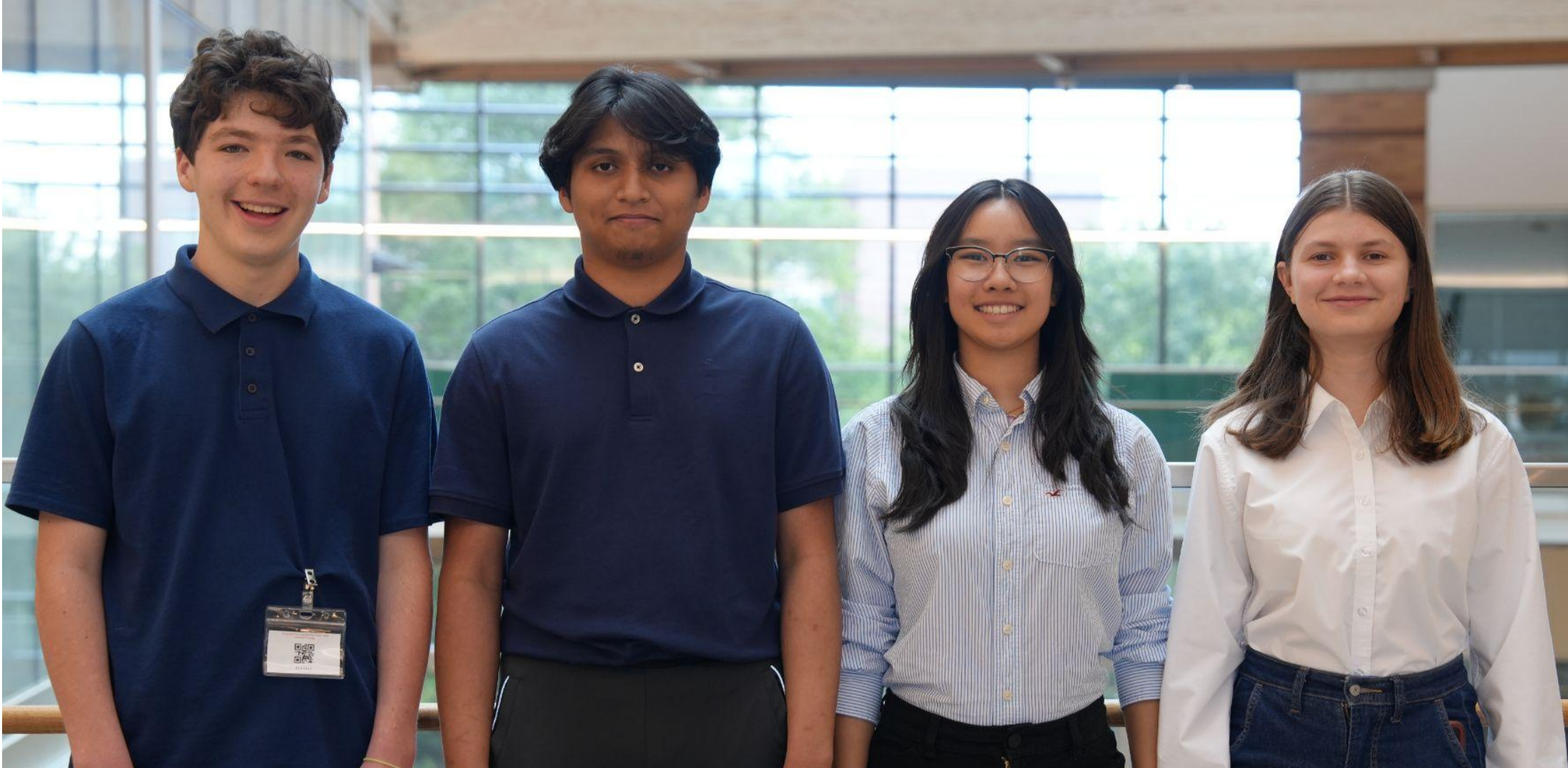
- **Over 500,000** people use ASL.
- **50:1** ASL users to interpreters.
- **90%** of deaf children are born to hearing parents who often have to **learn ASL**.
- **Communication barrier** prevents inclusivity in discussion spaces.

THE SOLUTION



SignScribe **translates live speech** into ASL fingerspelling and is a portable interpreter and learning tool for ASL. It differs from other products as it offers **2D and mechanical simulations** with **user customizations**.

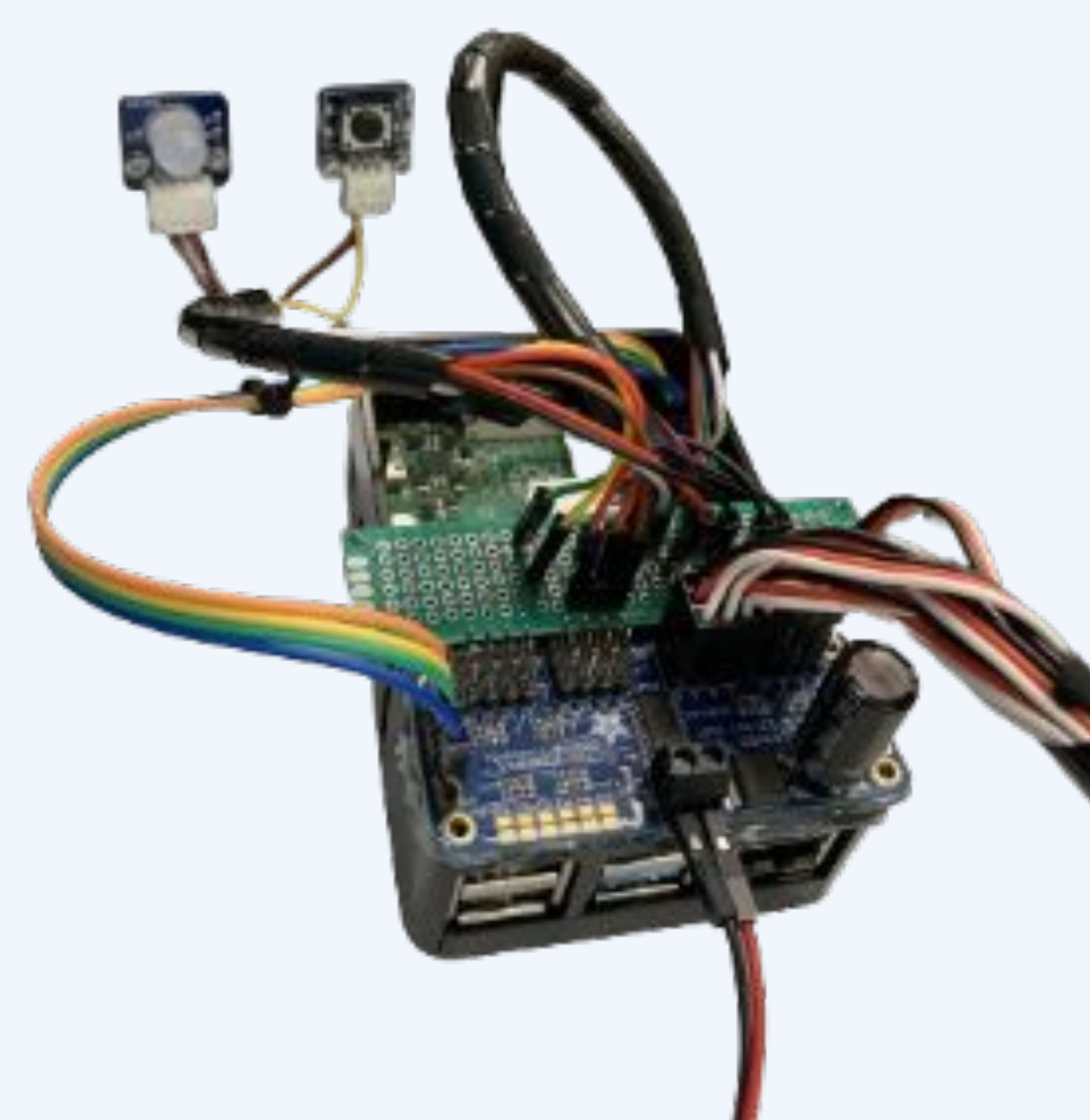
THE TEAM



Allison Andreyev: Project Leader, Software
Yenni Do: Hardware/CAD/Electronics
Sakib Niaz: Software/GUI
Elliott Owens: Computer Engineer/Electronics

THE TECHNOLOGY

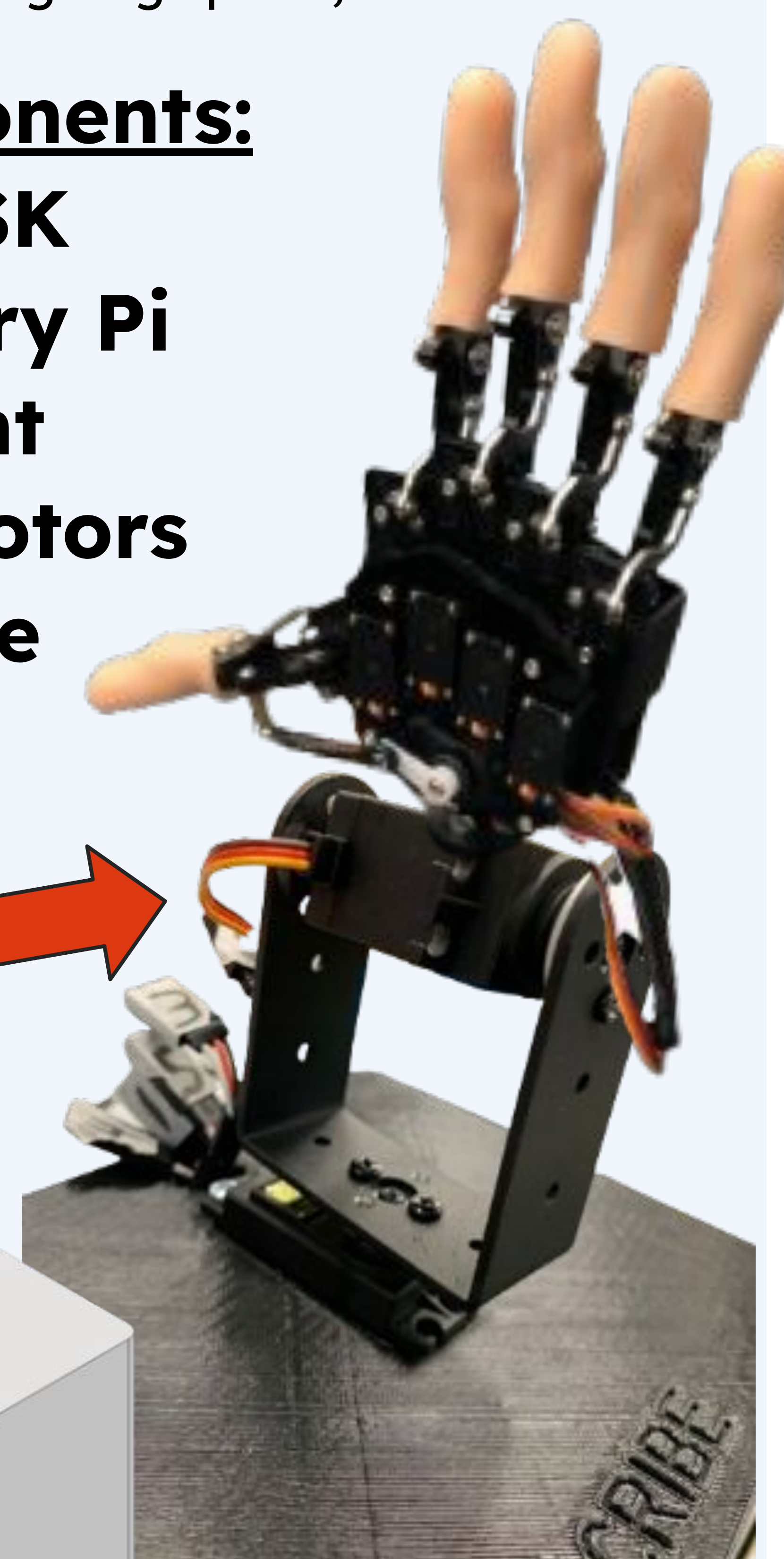
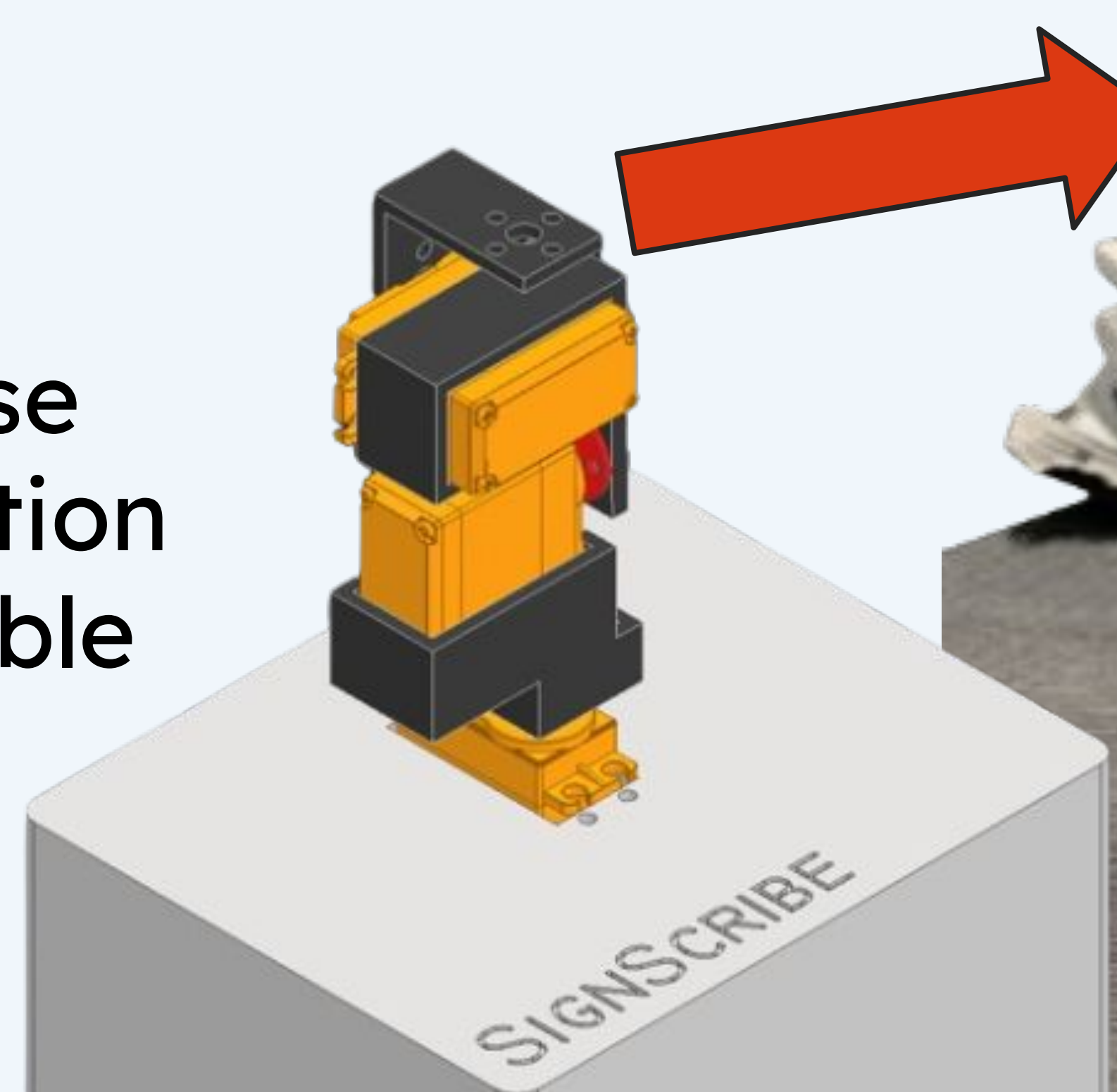
SignScribe **translates speech into text** using LLM called Vosk and then uses a Raspberry Pi **microcomputer** to control servos on each finger and the wrist. It also offers **customization through our configuration file**, so users can decide what words to censor, whether to auto launch the GUI or autosave generated transcript, change the signing speed, and more.



Key Components:

- **LLM VOSK**
- **Raspberry Pi**
- **LED Light**
- **Servo Motors**
- **CAD Base**

Our configuration file features **6 customizable features**. We plan to increase the customization options available as we further work on the prototype.

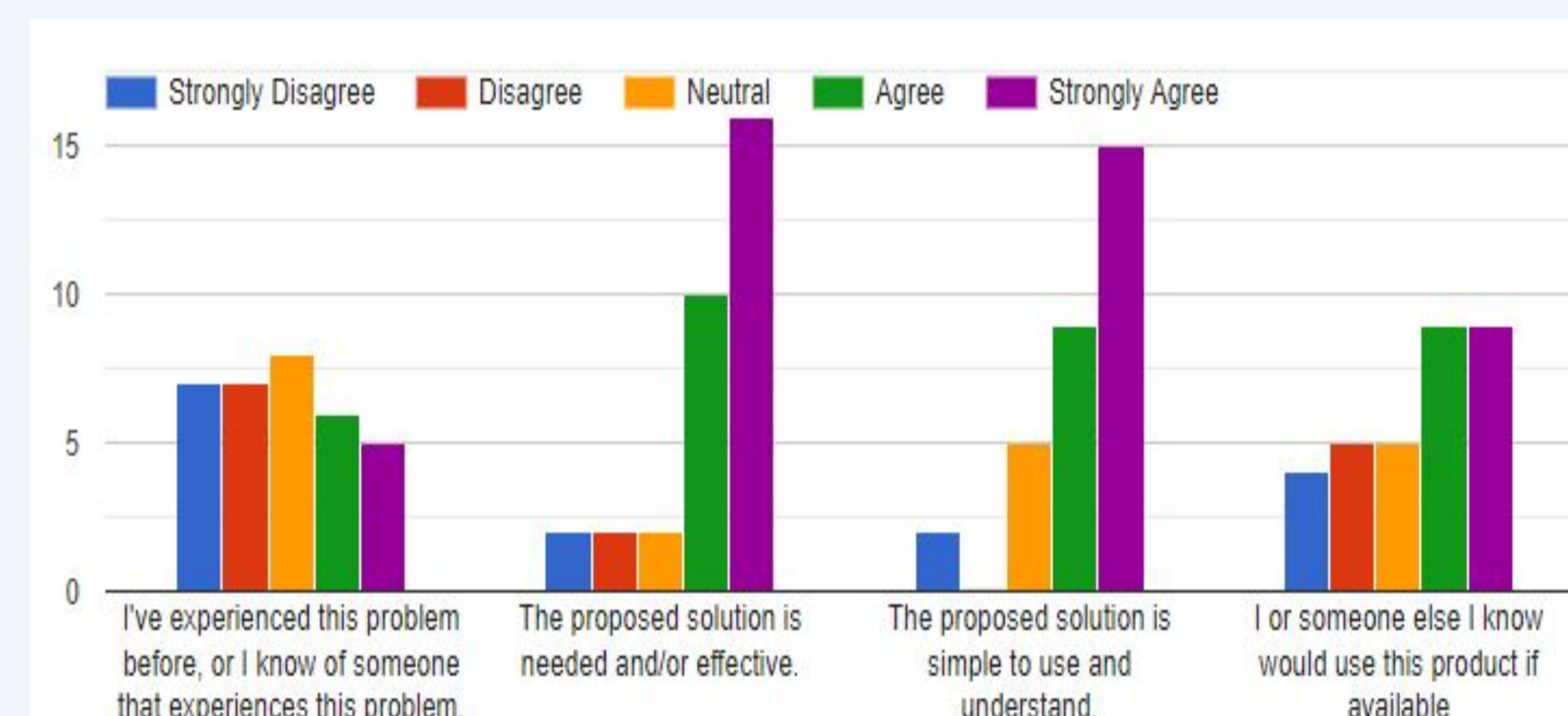


OTHER APPLICATIONS

- Courtrooms
- Schools, as a **learning tool**
- Political speeches, rallies
- Business meetings
- Conferences
- News streaming

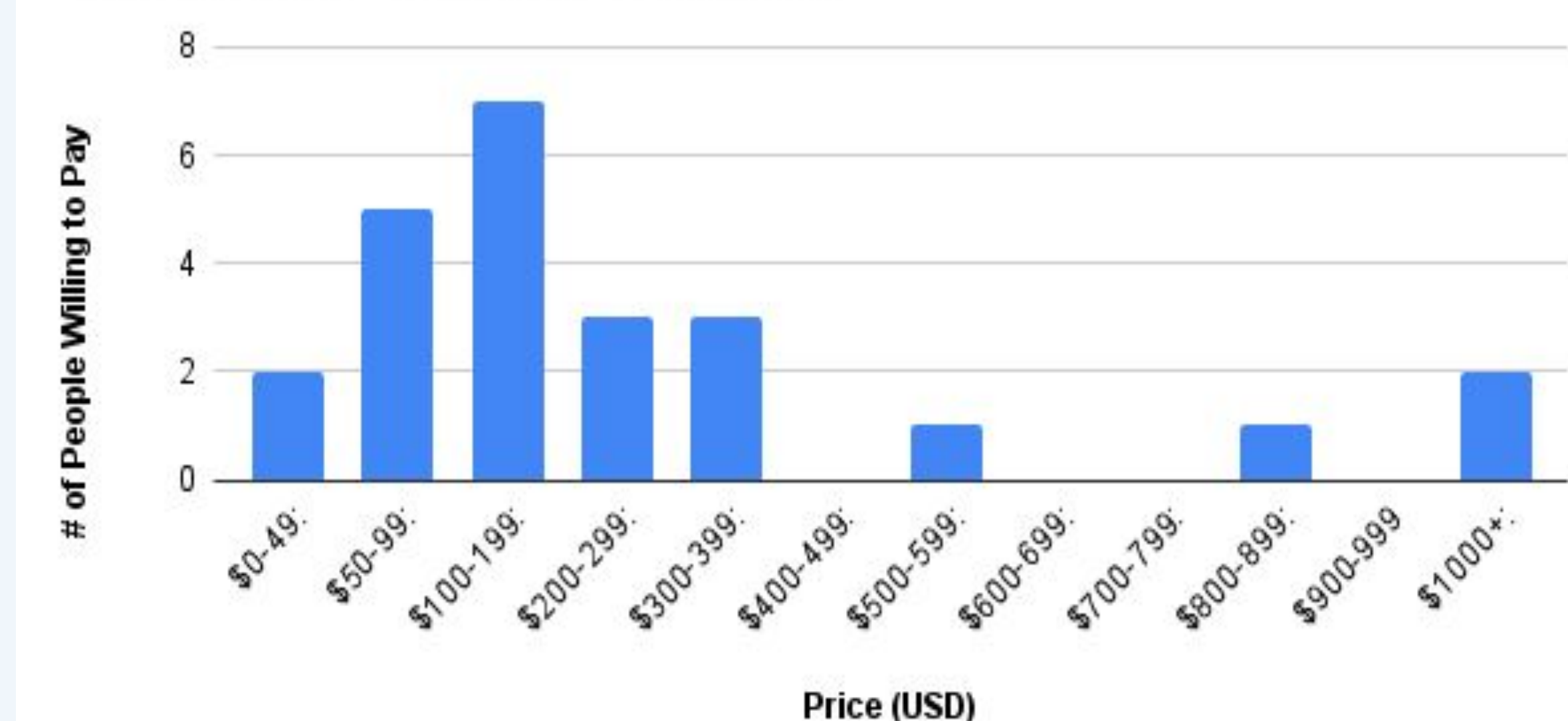


VALIDATION

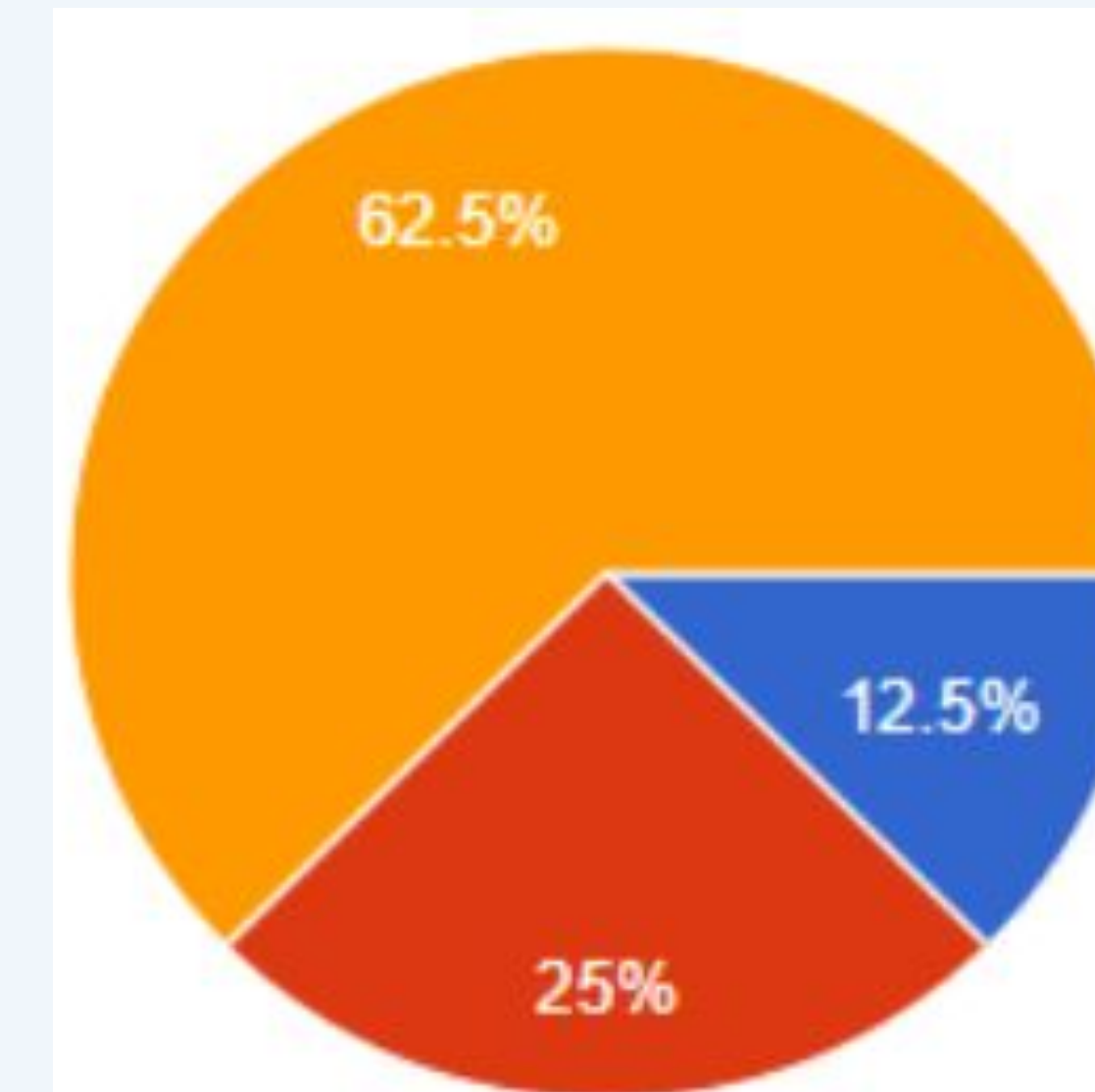


Over 75% of respondents said our **product is needed and effective** along with it being **simple to use and understand**.

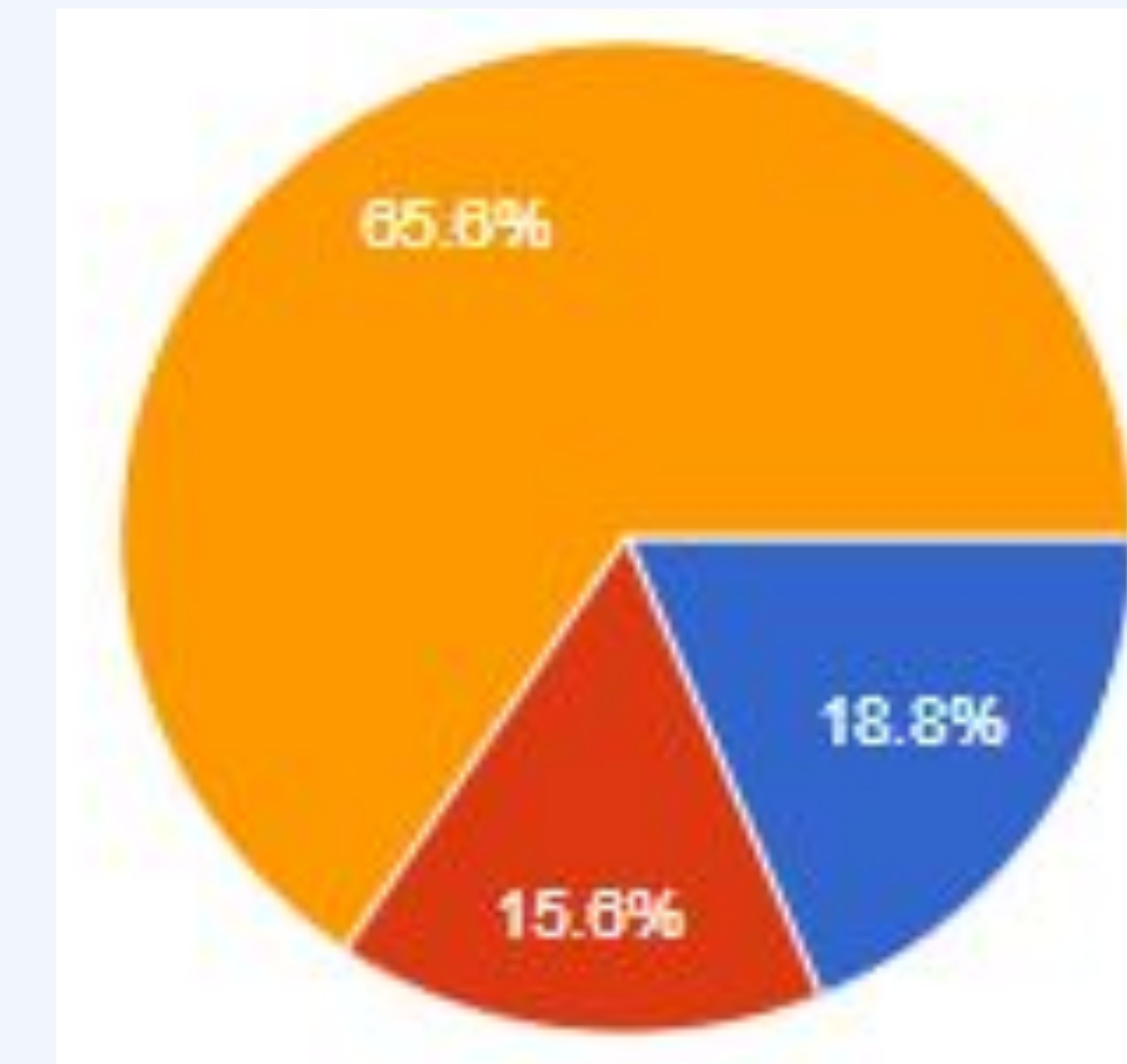
SignScribe Potential Price Data



On average, respondents said they would pay about **\$270** for this product.



Over 60% of those polls said they are now **interested in learning ASL** and **would use SignScribe** as a tool when doing so.



Over 60% of respondents said they prefer using **both the GUI and the mechanical hand** when using the product.